

Q-RAN **(ARIA)**

Yaml File Configuration



Q_RAN_Release_1.0

Overview

This document explains the structure and parameters of the Q-RAN YAML configuration files for both L3 and L2 components. Each section controls a specific part of the system—such as node identity, interface bindings, PLMN setup, logging, and cryptographic features.

The tables below list all configuration parameters along with their descriptions to help you set up and deploy Q-RAN nodes effectively

Q-RAN L3

ReleaseVersion

Parameter	Description
ReleaseVersion	Version identifier for the CU YAML configuration

gNodeB-CU

Parameter	Description
gNBName	Human-readable name for the L3 node
gNBId	Unique identifier for the gNB (in hex)
trackingAreaCode	TAC used to identify the tracking area
nRCellIdentity	NR Cell ID used by the L3 for this gNB

ngapConfig

Parameter	Description
n2IpAddress	Local L3 IP address for N2 (control-plane) interface
n3IpAddress	Local L3 IP address for N3 (user-plane) interface
amfN2IpAddress	IP address of the AMF (Access and Mobility Mgmt Function)
sctpRxStream	Number of SCTP receive streams
sctpTxStream	Number of SCTP transmit streams

f1apConfig

Parameter	Description
localIpAddress	L3's local IP for F1-C (control plane)
localPort	Port on which L3 listens for F1-C messages
duIpAddress	DU's IP address for F1-C
duPort	Port on which DU listens for F1-C messages

plmnConfig

Parameter	Description
mcc	Mobile Country Code
mnc	Mobile Network Code
mncLength	Length of the MNC (2 or 3 digits)
S-NSSAIGroup	
sst	Slice/Service Type (e.g., 1 = eMBB)
sd	Slice Differentiator (3-byte hex value)

cuConfig

Parameter	Description
sdapMode	Set to "disable" or "enable" for SDAP
isolatedCores	Comma-separated list of logical CPU core IDs to isolate L3 threads

logConfig

Parameter	Description
verticalLevel	Global log level (LOG_DEBUG, LOG_INFO, LOG_WARN, LOG_NONE)
Horizontal_Level.PDCP	Enable logging for PDCP module
Horizontal_Level.SCTP	Enable logging for SCTP module
Horizontal_Level.F1AP	Enable logging for F1AP module
Horizontal_Level.RRC	Enable logging for RRC module
Horizontal_Level.NGAP	Enable logging for NGAP module
Horizontal_Level.GTPU	Enable logging for GTP-U module
Horizontal_Level.CU_APP	Enable logging for L3 application logic
Horizontal_Level.ARIA	Enable logging for ARIA crypto module
Horizontal_Level.SDAP	Enable logging for SDAP module
Horizontal_Level.DTLS	Enable logging for DTLS module
printTimeStamp	Include timestamp in log entries

generateLogFile

Parameter	Description
generateLogFile	Whether to generate runtime log files (true/false)

logFileConfig

Parameter	Description
verticalLevel	File log level (LOG_DEBUG, LOG_INFO, LOG_WARN, LOG_NONE)
Horizontal_Level.PDCP	Enable logging to file for PDCP module
Horizontal_Level.SCTP	Enable logging to file for SCTP module
Horizontal_Level.F1AP	Enable logging to file for F1AP module
Horizontal_Level.RRC	Enable logging to file for RRC module
Horizontal_Level.NGAP	Enable logging to file for NGAP module
Horizontal_Level.GTPU	Enable logging to file for GTP-U module
Horizontal_Level.CU_APP	Enable logging to file for CU application logic
Horizontal_Level.ARIA	Enable logging to file for ARIA crypto module
Horizontal_Level.SDAP	Enable logging to file for SDAP module
Horizontal_Level.DTLS	Enable logging to file for DTLS module
Horizontal_Level.printTimeStamp	Include timestamps in file log entries

Q-RAN L2

ReleaseVersion

Parameter	Description
ReleaseVersion	Version of the DU YAML configuration

gNodeB-DU

Parameter	Description
gNBName	Name of the DU node
gNBId	Common gNB ID shared with CU
gNBDUIId	Unique identifier for this DU
trackingAreaCode	Tracking Area Code
nRCellIdentity	NR Cell Identity for the DU

plmnConfig

Parameter	Description
mcc	Mobile Country Code
mnc	Mobile Network Code
mncLength	Length of the MNC (2 or 3 digits)
S-NSSAIGroup	
sst	Slice/Service Type (e.g., 1 = eMBB)
sd	Slice Differentiator (3-byte hex value)

flapInterface

Parameter	Description
numComponentCarriers	Number of carriers supported
culpAddress	IP of CU for F1 interface
cuPort	Port of CU for F1 interface
dulpAddress	IP of DU
duPort	Port of DU for F1 interface

duConfig

Parameter	Description
duCores	List of CPU cores used by DU
puschAntennaPort	PUSCH antenna ports
pdschAntennaPortXP	PDSCH antenna ports per X-PHY
csirsEnabled	CSI-RS enable flag (1 or 0)
srsEnabled	SRS enable flag (1 or 0)
sib1TimeDomainAllocation	Time domain resource for SIB1

downlinkConfigCommon

Parameter	Description
absoluteFrequencySSB	SSB frequency (in kHz)
dlfrequencyBand	Frequency band for downlink
absoluteFrequencyPointA	Absolute frequency of point A
scs-SpecificCarrierList	Placeholder (not expanded)
dlsubcarrierSpacing	Subcarrier spacing for DL
dlcarrierBandwidth	DL carrier bandwidth
lldownlinkOffsetToCarrier	Offset to carrier in DL

initialDownlinkBWPParams

Parameter	Description
subcarrierSpacing	SCS for BWP
locationAndBandwidth	Location and bandwidth index
searchSpaceZero	Search space ID
controlResourceSetZero	CORESET ID

uplinkConfigCommon

Parameter	Description
ulfrequencyBand	UL frequency band
scsSpecificCarrierList	Placeholder (not expanded)
ulsubcarrierSpacing	UL subcarrier spacing
ulcarrierBandwidth	UL carrier bandwidth
uloffsetToCarrier	Offset in UL
maximumTransmissionPower	Max UE TX power (in dBm)

initialUplinkBWPParams

Parameter	Description
subcarrierSpacing	SCS for initial uplink BWP
locationAndBandwidth	Location and bandwidth index

RACHConfigCommon

Parameter	Description
prachConfigIndex	PRACH index
prachMsg1FDM	FDM for Msg1
prachMsg1FrequencyStart	Frequency offset for Msg1
prachRootSequenceIndexPR	PR selector index
prachRootSequenceIndexValue	Root sequence value
msg1SubcarrierSpacing	Subcarrier spacing for Msg1
zeroCorrelationZoneConfig	ZCZ configuration
preambleReceivedTargetPower	Target power for PRACH preamble
preambleTransmissionMax	Max transmission attempts
ssbPerRACHOccasion	Number of SSBs per RACH
cbPreamblesPerSSB	Number of contention-based preambles

Parameter	Description
raContentionResolutionTimer	Contention resolution timer
msg3DeltaPreamble	Offset for Msg3
p0NominalWithGrant	Power offset with grant
rsrpThresholdSSB	RSRP threshold for SSB
restrictedSetConfig	RACH restricted set config
powerRampingStep	Power ramping step
raResponseWindow	Response window in ms

pucchConfigCommon

Parameter	Description
pucchGroupHopping	Group hopping enabled/disabled
hoppingID	Hopping ID
p0Nominal	Nominal P0
ssbPositionsInBurstBitmapType	Type of bitmap for SSB burst
ssbPositionsInBurst	SSB position in burst
ssbPeriodicityServingCell	Periodicity of SSB
dmrsTypeAPosition	DMRS type-A position
subcarrierSpacing	SCS used in PUCCH

servingCellConfigCommon

Parameter	Description
physicalCellIdentity	PCI used for the DU cell

pdccchSIB1Configs

Parameter	Description
controlResourceSetZero	CORESET ID for SIB1
searchSpaceZero	Search space for SIB1

tddULDLConfigCommon

Parameter	Description
ssPBCHBlockPower	Transmit power for SS/PBCH block
nrofDownlinkSlots	Number of DL slots in TDD frame
nrofDownlinkSymbols	Number of DL symbols in partial slot
nrofUplinkSlots	Number of UL slots
nrofUplinkSymbols	Number of UL symbols
referenceSubcarrierSpacing	Reference SCS
dlULTransmissionPeriodicity	DL/UL switching periodicity (ms)

L1ConfigList

Parameter	Description
numComponentCarriers	Number of carriers in L1
rxThreadCore	CPU core for RX thread
txThreadCore	CPU core for TX thread
txAmplifierBackoff	TX power backoff
prachDtxThreshold	DTX threshold for PRACH
pucch0DtxThreshold	DTX threshold for PUCCH0
puschDtxThreshold	DTX threshold for PUSCH
phaseCompensationEnabled	Enable phase compensation

RUConfigList

Parameter	Description
slotAhead	Number of slots ahead for RU scheduling
numTransmitAntennas	Number of TX antennas
numReceiveAntennas	Number of RX antennas
ruThreadCore	Core used by RU threads
transmitAttenuation	TX attenuation
receiveAttenuation	RX attenuation

Parameter	Description
frequencyBands	Operating frequency band
maxPdschReferenceSignalPower	Max RS power for PDSCH
maxRxGain	Max RX gain
precodingEnabled	Whether precoding is enabled
beamformingWeights	List of beamforming weights

FHI7p2Config

Parameter	Description
mtuForRu	MTU size for RU interface
duAddressList	MAC addresses of DU
ruAddressList	MAC addresses of RU
dppdkDeviceList	PCI IDs of DPDK interfaces
dppdkCoreIsolation	Core ID for isolating DPDK threads
fhiCoreIsolation	Core ID used for FHI
fhiWorkerCoreIsolation	List of worker core IDs for FHI

fhTimingConfig

Parameter	Description
Tadv_CP_DL, T2a_*, T1a_*, Ta3_Min, Ta4_Max	Timing offsets and alignment for IQ
prachTimingConfig.eAxCOffset	PRACH eAxC offset
prachTimingConfig.kBar	kBar value
ruTimingConfig.iqBitWidth	Bit width for IQ samples
ruTimingConfig.fftSize	FFT size
ruTimingConfig.prachIqBitWidth	Bit width for PRACH IQ

L1StatsUpdateInterval

Parameter	Description
L1StatsUpdateInterval	Time (in seconds) for L1 stats update

generateLogFile

Parameter	Description
generateLogFile	Whether to generate runtime log files (true/false)

logConfig

Parameter	Description
verticalLevel	Console log level
horizontalLevel.SCTP	Enable SCTP module logging
horizontalLevel.F1AP	Enable F1AP logging
horizontalLevel.GTPU	Enable GTP-U logging
horizontalLevel.RLC	Enable RLC logging
horizontalLevel.MAC	Enable MAC logging
horizontalLevel.PHY	Enable PHY logging
horizontalLevel.ARIA	Enable ARIA logging
horizontalLevel.GNB_APP	Enable GNB application logic logging
horizontalLevel.DTLS	Enable DTLS logging
printTimeStamp	Print timestamps in console logs

logsFileConfig

Parameter	Description
verticalLevel	File log level
horizontalLevel.SCTP	Enable SCTP file logging
horizontalLevel.F1AP	Enable F1AP file logging
horizontalLevel.GTPU	Enable GTP-U file logging
horizontalLevel.RLC	Enable RLC file logging
horizontalLevel.MAC	Enable MAC file logging
horizontalLevel.PHY	Enable PHY file logging
horizontalLevel.ARIA	Enable ARIA file logging
horizontalLevel.GNB_APP	Enable GNB application file logging
horizontalLevel.DTLS	Enable DTLS file logging
printTimeStamp	Print timestamps in log files